

A Sharp Best-Sequence Benchmark Bound for Liquid Welfare

Partial Progress on Stronger Benchmarks in Sequential Budgeted First-Price Auctions

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Abstract

Fikioris and Tardos ask whether stronger natural utility benchmarks, such as richer value-to-bid maps or stochastic-value benchmarks, can improve the liquid-welfare guarantees known for sequential budgeted first-price auctions. This note gives a complete answer for one particularly strong benchmark: the best budget-feasible sequence of bids in hindsight. If every bidder's realized utility is within a factor $\gamma \geq 1$, up to additive regret r_i , of the utility of her best budget-feasible bid sequence against the realized competing bids, then

$$\text{LW} \geq \frac{\text{LW}^* - n}{\max\{\gamma, 2\}} - \sum_i r_i.$$

The factor $\max\{\gamma, 2\}$ is optimal up to additive lower-order terms, by lower-bound examples already implicit in the work of Fikioris and Tardos and reproduced here.

Thus the note is a solution for the best-sequence benchmark. As an answer to the broader open direction, it should be read as partial progress: it also gives the same guarantee for the unrestricted all-Lipschitz value-to-bid benchmark on sample paths with distinct values, hence almost surely for independent atomless values, but it does not settle benchmarks with a Lipschitz constant fixed in advance, nor does it prove that stochasticity alone improves the original fixed-shading benchmark.

1 Introduction and status of the result

Sequential budgeted first-price auctions are a basic model for repeated ad-auction markets. There are T rounds, one item is sold in each round, and each bidder has a hard budget over the full sequence. Fikioris and Tardos [1, 2] prove liquid-welfare guarantees assuming that each bidder obtains utility comparable to the best fixed multiplicative shading of her values in hindsight. For additive values their bound has competitive factor

$$\gamma + \frac{1}{2} + O(\gamma^{-1}),$$

which is about 2.41 in the no-regret case $\gamma = 1$. They also prove lower bounds of 2 for $\gamma = 1$ and γ for general γ , and these lower bounds already hold even relative to the stronger benchmark of the best sequence of bids.

The natural gap is therefore the following. The known lower bounds say that no benchmark of this kind can force a factor better than $\max\{\gamma, 2\}$, even if bidders are competitive against the best sequence of bids. The fixed-shading analysis, however, only proves the larger factor $\gamma + 1/2 + O(1/\gamma)$. This note supplies the missing upper bound for the best-sequence benchmark.

What is solved. For the benchmark consisting of all budget-feasible bid sequences in hindsight, the exact asymptotic liquid-welfare factor is

$$\boxed{\max\{\gamma, 2\}}.$$

In particular, no-regret relative to the best sequence of bids implies a sharp factor-2 liquid-welfare guarantee, up to the standard additive loss from indivisible items and regret.

What remains partial. This is not a full solution of every possible reading of the broader open question. The argument applies to the unrestricted class of all Lipschitz value-to-bid maps because, on a finite sample path with distinct realized values, any bid sequence can be interpolated by a Lipschitz map with some finite sample-dependent Lipschitz constant. It does not apply to a class of L -Lipschitz maps when L is fixed before the sample is observed. It also does not show that stochasticity alone improves the fixed-shading benchmark.

2 Model

There are bidders $i \in [n]$ and rounds $t \in [T]$. In round t , one item is sold by a first-price auction. Bidder i has value $v_{it} \in [0, 1]$ for that item and has a hard budget $B_i > 0$ over all rounds. Let $b_{it} \geq 0$ be the realized bid of bidder i in round t , and set

$$q_t := \max_j b_{jt}, \quad h_{it} := \max_{j \neq i} b_{jt}.$$

Thus q_t is the winning first-price payment in round t , and h_{it} is the highest competing bid faced by bidder i . Ties are broken by an arbitrary fixed rule. All deviations in the proof bid $h_{it} + \eta$, so tie-breaking never affects the argument.

Let X_i be the set of rounds won by bidder i in the realized play. Define

$$V_i := \sum_{t \in X_i} v_{it}, \quad P_i := \sum_{t \in X_i} q_t, \quad U_i := V_i - P_i.$$

We assume the realized play respects budgets, so $P_i \leq B_i$ for every bidder. The realized liquid welfare is

$$\text{LW} := \sum_i \text{LW}_i, \quad \text{LW}_i := \min\{B_i, V_i\}.$$

The optimal liquid welfare is

$$\text{LW}^* := \max_{(A_i)_i} \sum_i \min \left\{ B_i, \sum_{t \in A_i} v_{it} \right\},$$

where the maximum is over all allocations of the T items to the bidders.

2.1 The best-sequence benchmark

Fix the realized bids of all bidders except bidder i . A counterfactual bid sequence $a = (a_t)_{t=1}^T$ is feasible for bidder i if $a_t \in [0, 1]$ for every t and

$$\sum_{t: a_t > h_{it}} a_t \leq B_i.$$

This is exactly the budget used by the sequence in the rounds it wins, since the auction is first-price. The utility of such a sequence is

$$U_i(a) := \sum_{t: a_t > h_{it}} (v_{it} - a_t).$$

Define

$$\text{SEQ}_i := \sup_a U_i(a),$$

where the supremum is over feasible bid sequences. The zero bid sequence is feasible, so $\text{SEQ}_i \geq 0$.

We say that bidder i is (γ, r_i) -competitive against the best-sequence benchmark if

$$U_i \geq \frac{1}{\gamma} \text{SEQ}_i - r_i, \quad \gamma \geq 1, \quad (1)$$

where $r_i \geq 0$ is an additive regret term. The usual no-regret case is $\gamma = 1$ and $r_i = o(T)$.

3 The upper bound

The proof has three ingredients. The first trims an optimal liquid-welfare allocation so that the selected value for each bidder is at most her budget.

Lemma 1 (Truncating an optimal allocation). *Let $(A_i)_i$ be any allocation, and set $z_i = \sum_{t \in A_i} v_{it}$. For each bidder i there is a subset $S_i \subseteq A_i$ such that, writing*

$$c_i := \sum_{t \in S_i} v_{it},$$

we have

$$c_i \leq B_i \quad \text{and} \quad c_i \geq \min\{B_i, z_i\} - 1.$$

Consequently, if $(A_i)_i$ is optimal, then the sets S_i can be chosen disjointly and satisfy

$$\sum_i c_i \geq \text{LW}^* - n. \quad (2)$$

Proof. If $z_i \leq B_i$, take $S_i = A_i$. If $z_i > B_i$, scan the items of A_i in an arbitrary order and keep adding items as long as the selected total value remains at most B_i . The resulting selected value c_i is at most B_i . If $c_i < B_i - 1$, then every unselected item of A_i would have value strictly larger than $B_i - c_i > 1$, contradicting $v_{it} \leq 1$. Hence $c_i \geq B_i - 1 = \min\{B_i, z_i\} - 1$ in this case. $\square$

The second ingredient converts the truncated optimum into the sum of actual payments and benchmark deviation utilities.

Lemma 2 (Payments plus sequence surplus). *Fix disjoint sets S_i such that $c_i := \sum_{t \in S_i} v_{it} \leq B_i$ for every bidder. For $\eta > 0$, define*

$$s_i(\eta) := \sum_{t \in S_i} (v_{it} - h_{it} - \eta)_+.$$

Then $s_i(\eta) \leq \text{SEQ}_i$ for every i , and

$$\sum_i P_i + \sum_i s_i(\eta) \geq \sum_i c_i - \eta T. \quad (3)$$

Proof. For bidder i , consider the bid sequence that bids $h_{it} + \eta$ on the rounds $t \in S_i$ for which $h_{it} + \eta \leq v_{it}$, and bids 0 otherwise. Its total payment is at most

$$\sum_{t \in S_i: h_{it} + \eta \leq v_{it}} (h_{it} + \eta) \leq \sum_{t \in S_i} v_{it} = c_i \leq B_i,$$

so it is feasible. Its utility is exactly $s_i(\eta)$, proving $s_i(\eta) \leq \text{SEQ}_i$.

For every selected item $t \in \bigcup_i S_i$, let $i(t)$ be the unique bidder with $t \in S_{i(t)}$. Since $h_{i(t)t} \leq q_t$, and since the selected sets are disjoint,

$$\sum_{t \in \bigcup_i S_i} h_{i(t)t} \leq \sum_{t=1}^T q_t = \sum_i P_i.$$

For each selected item,

$$h_{i(t)t} + (v_{i(t)t} - h_{i(t)t} - \eta)_+ \geq v_{i(t)t} - \eta.$$

Adding over selected items gives (3). $\square$

The final ingredient is the only algebraic point at which the factor $\max\{\gamma, 2\}$ appears.

Lemma 3 (Liquid-welfare algebra). *Let $0 \leq p \leq B$, $0 \leq u \leq B$, $\gamma \geq 1$, $r \geq 0$, and $\beta := \max\{\gamma, 2\}$. If $U \geq u/\gamma - r$, then*

$$\min\{B, p + U\} \geq \frac{p + u}{\beta} - r. \quad (4)$$

Proof. Because $p \leq B$,

$$\min\{B, p + U\} = p + \min\{B - p, U\}.$$

From $U \geq u/\gamma - r$ we get

$$\min\{B - p, U\} \geq \min\{B - p, u/\gamma\} - r.$$

It remains to prove

$$p + \min\{B - p, u/\gamma\} \geq \frac{p + u}{\beta}.$$

If $u \leq \gamma(B - p)$, then the left-hand side is $p + u/\gamma$. When $\gamma \geq 2$ this is at least $(p + u)/\gamma = (p + u)/\beta$, and when $\gamma \leq 2$ it is at least $(p + u)/2 = (p + u)/\beta$. If $u > \gamma(B - p)$, then the left-hand side is B . Since $p \leq B$, $u \leq B$, and $\beta \geq 2$, we have $B \geq (p + u)/2 \geq (p + u)/\beta$. $\square$

Theorem 1 (Sharp best-sequence liquid-welfare guarantee). *Assume every bidder is (γ, r_i) -competitive against the best-sequence benchmark in the sense of (1). Let*

$$R := \sum_i r_i, \quad \beta := \max\{\gamma, 2\}.$$

Then

$$\text{LW} \geq \frac{\text{LW}^* - n}{\beta} - R. \quad (5)$$

Equivalently,

$$\text{LW}^* \leq \beta(\text{LW} + R) + n.$$

If item values are bounded by Δ instead of 1, the additive term n becomes $n\Delta$. If items are divisible so that the truncation in Lemma 1 loses no value, the additive term is 0.

Proof. Let $(A_i)_i$ be an optimal allocation for LW^* , and choose disjoint subsets $S_i \subseteq A_i$ as in Lemma 1. Write $c_i = \sum_{t \in S_i} v_{it}$. Fix $\eta > 0$ and put

$$u_i := s_i(\eta) = \sum_{t \in S_i} (v_{it} - h_{it} - \eta)_+.$$

By Lemma 2, $u_i \leq \text{SEQ}_i$. Also $0 \leq u_i \leq c_i \leq B_i$. The competitiveness assumption gives

$$U_i \geq \frac{1}{\gamma} \text{SEQ}_i - r_i \geq \frac{u_i}{\gamma} - r_i.$$

Apply Lemma 3 with $p = P_i$, $B = B_i$, $U = U_i$, and $u = u_i$. Since $V_i = P_i + U_i$,

$$\text{LW}_i = \min\{B_i, V_i\} = \min\{B_i, P_i + U_i\} \geq \frac{P_i + u_i}{\beta} - r_i.$$

Summing over bidders and using Lemma 2,

$$\text{LW} \geq \frac{\sum_i P_i + \sum_i s_i(\eta)}{\beta} - R \geq \frac{\sum_i c_i - \eta T}{\beta} - R.$$

By Lemma 1, $\sum_i c_i \geq \text{LW}^* - n$. Letting $\eta \downarrow 0$ proves (5). $\square$

4 Tightness for the best-sequence benchmark

The next examples are essentially the lower bounds of Fikioris and Tardos [2]. They are included to make the sharpness statement self-contained. Their important feature here is that the bidders are already competitive against the best sequence of bids, not merely against the best fixed shading.

Proposition 1 (The factor 2 is unavoidable). *For every sufficiently large T and every ε with $1/T \leq \varepsilon \leq 1$, there is a two-bidder instance in which both bidders have competitive ratio 1 and additive regret at most 1 against the best-sequence benchmark, while*

$$\frac{\text{LW}}{\text{LW}^*} \leq \frac{1}{2 - \varepsilon - 1/T}.$$

Consequently no asymptotic guarantee better than factor 2 is possible for no-regret against the best sequence of bids.

Proof. Bidder 1 has value $v_{1t} = 1$ and budget $B_1 = T\varepsilon$. Bidder 2 has value $v_{2t} = \varepsilon$ and budget $B_2 = T\varepsilon$. In every round bidder 1 bids ε , bidder 2 bids $\varepsilon - 1/T$, and bidder 1 wins. The realized liquid welfare is

$$\text{LW} = \min\{T\varepsilon, T\} = T\varepsilon.$$

An allocation giving bidder 1 the first $\lceil T\varepsilon \rceil$ items and bidder 2 the remaining items has liquid welfare at least

$$T\varepsilon + (T - \lceil T\varepsilon \rceil)\varepsilon \geq T(2\varepsilon - \varepsilon^2) - \varepsilon.$$

Thus

$$\frac{\text{LW}}{\text{LW}^*} \leq \frac{T\varepsilon}{T(2\varepsilon - \varepsilon^2) - \varepsilon} = \frac{1}{2 - \varepsilon - 1/T}.$$

Bidder 2 cannot obtain positive utility by winning, because she would need to bid strictly above ε against bidder 1 and her value is ε . Hence her best sequence utility is 0, equal to her realized utility. Bidder 1's realized utility is $T(1 - \varepsilon)$. Her best sequence can win every item at price just above $\varepsilon - 1/T$, obtaining utility arbitrarily close to $T(1 - \varepsilon) + 1$. Hence her additive regret is at most 1. $\square$

Proposition 2 (The factor γ is unavoidable). *For every fixed $\gamma \geq 1$ and all sufficiently large T , there is a two-bidder instance with additive regret at most 1 against the best-sequence benchmark in which both bidders are γ -competitive and*

$$\frac{\text{LW}}{\text{LW}^*} \leq \frac{1}{\gamma} + o(1).$$

Consequently the factor in Theorem 1 is asymptotically optimal for every $\gamma \geq 1$.

Proof. Assume for notational simplicity that T/γ is an integer; rounding changes only $o(T)$ terms. Budgets are nonbinding. Bidder 1 has value $v_{1t} = 1$ in every round. Bidder 2 has value $v_{2t} = \varepsilon := 1/T$ in every round. The optimal allocation gives all items to bidder 1, so $\text{LW}^* = T$.

For the first T/γ rounds, bidder 1 bids ε and bidder 2 bids 0. For the remaining rounds, bidder 1 bids 0 and bidder 2 bids ε^2 . Therefore bidder 1 wins the first T/γ items and bidder 2 wins the rest. The realized liquid welfare is

$$\text{LW} = \frac{T}{\gamma} + \varepsilon T \left(1 - \frac{1}{\gamma}\right) = \frac{T}{\gamma} + 1 - \frac{1}{\gamma},$$

which is $(1/\gamma) + o(1)$ times $\text{LW}^* = T$.

Bidder 1's realized utility is $(1 - \varepsilon)T/\gamma$. Her best sequence can win the first T/γ items at price arbitrarily close to 0, and the remaining items at price arbitrarily close to ε^2 , for utility arbitrarily close to

$$\frac{T}{\gamma} + (1 - \varepsilon^2)T \left(1 - \frac{1}{\gamma}\right) = T \left(1 - \varepsilon^2 \left(1 - \frac{1}{\gamma}\right)\right).$$

Dividing this benchmark by γ exceeds her realized utility by less than 1. Bidder 2's realized utility is

$$(\varepsilon - \varepsilon^2)T \left(1 - \frac{1}{\gamma}\right),$$

whereas her best sequence utility is

$$\varepsilon T \left(1 - \frac{1}{\gamma}\right).$$

Thus her ordinary additive regret is $\varepsilon^2 T(1 - 1/\gamma) \leq 1$, and a fortiori she is γ -competitive with additive regret at most 1. $\square$

Theorem 1 and Propositions 1–2 give the exact asymptotic answer for the best-sequence benchmark:

$$\alpha_{\text{seq}}(\gamma) = \max\{\gamma, 2\}.$$

5 Unrestricted Lipschitz value-to-bid benchmarks

Let $\mathcal{L}_{\text{all}}$ denote the class of all Lipschitz functions $f : [0, 1] \rightarrow [0, 1]$, with no Lipschitz constant fixed in advance. For bidder i , let LIP_i be the supremum of the utility obtained by bidding $f(v_{it})$ in every round, over all $f \in \mathcal{L}_{\text{all}}$ whose induced bids are budget-feasible:

$$\sum_{t: f(v_{it}) > h_{it}} f(v_{it}) \leq B_i.$$

Proposition 3 (Equivalence on distinct sample paths). *If $v_{i1}, \dots, v_{iT}$ are pairwise distinct, then*

$$\text{LIP}_i = \text{SEQ}_i.$$

Proof. Every Lipschitz rule f induces the feasible bid sequence $a_t = f(v_{it})$, so $\text{LIP}_i \leq \text{SEQ}_i$.

Conversely, fix any feasible bid sequence $a_1, \dots, a_T \in [0, 1]$. Sort the distinct realized values as

$$x_1 < x_2 < \dots < x_T,$$

where $x_k = v_{it_k}$, and prescribe $f(x_k) = a_{t_k}$. Define f by linear interpolation on each interval $[x_k, x_{k+1}]$, by the constant value a_{t_1} on $[0, x_1]$, and by the constant value a_{t_T} on $[x_T, 1]$. Then f maps $[0, 1]$ into $[0, 1]$ and has finite Lipschitz constant

$$\max_{1 \leq k < T} \frac{|a_{t_{k+1}} - a_{t_k}|}{x_{k+1} - x_k},$$

with the convention that the maximum is 0 when $T = 1$. The induced Lipschitz bidding rule exactly reproduces the sequence a_t on the realized rounds, hence has the same budget use and utility. Taking suprema over feasible sequences gives $\text{LIP}_i \geq \text{SEQ}_i$. $\square$

Corollary 1 (Atomless stochastic values). *Suppose that for each bidder i , the values $v_{i1}, \dots, v_{iT}$ are independent draws from atomless distributions. Then, with probability one, each bidder's realized values are pairwise distinct. On every such sample path, if every bidder is (γ, r_i) -competitive against the unrestricted all-Lipschitz benchmark $\mathcal{L}_{\text{all}}$, then*

$$\text{LW} \geq \frac{\text{LW}^* - n}{\max\{\gamma, 2\}} - \sum_i r_i.$$

Proof. Pairwise distinctness holds almost surely for finitely many independent draws from atomless distributions. Proposition 3 then identifies the all-Lipschitz benchmark with the best-sequence benchmark for each bidder, and Theorem 1 applies path by path. $\square$

The factor 2 in Corollary 1 cannot be improved in the no-regret case, even if one insists on atomless values. The following perturbation of Proposition 1 gives a self-contained statement.

Proposition 4 (Atomless factor-2 lower bound for no-regret). *There is a sequence of two-bidder atomless stochastic instances, with deterministic budget-feasible bidding rules, such that with probability one both bidders have additive regret at most 1 against the best-sequence benchmark, equivalently against the all-Lipschitz benchmark, while*

$$\frac{\text{LW}}{\text{LW}^*} \rightarrow \frac{1}{2}.$$

Proof. Let $\varepsilon = \varepsilon_T$ satisfy $\varepsilon \rightarrow 0$ and $T\varepsilon \rightarrow \infty$, for example $\varepsilon = T^{-1/2}$. Put $\delta = 1/(4T)$ and assume T is large enough that $\varepsilon > 2\delta$ and $\varepsilon \leq 1/2$. Bidder 1 has budget $B_1 = T\varepsilon$ and values drawn independently from the uniform distribution on $[1 - \delta, 1]$. Bidder 2 has budget $B_2 = T\varepsilon$ and values drawn independently from the uniform distribution on $[\varepsilon - \delta, \varepsilon]$. These distributions are atomless, so all realized values of each bidder are pairwise distinct with probability one.

In every round bidder 1 bids ε , bidder 2 bids $\varepsilon - 2\delta$, and bidder 1 wins. Since $\varepsilon \leq 1/2 < 1 - \delta$, bidder 1's total realized value exceeds B_1 , and therefore

$$\text{LW} = T\varepsilon.$$

For the optimum, give bidder 1 any

$$m := \left\lceil \frac{T\varepsilon}{1 - \delta} \right\rceil$$

items. Even in the worst realization, those m items have total value at least $m(1 - \delta) \geq T\varepsilon$, so bidder 1 contributes $T\varepsilon$ to liquid welfare. Give the remaining items to bidder 2. Their total value is at least $(T - m)(\varepsilon - \delta)$. Hence

$$\text{LW}^* \geq T\varepsilon + (T - m)(\varepsilon - \delta).$$

Because $m \leq T\varepsilon/(1 - \delta) + 1$, this implies

$$\frac{\text{LW}^*}{T\varepsilon} \geq 1 + \left(1 - \frac{\varepsilon}{1 - \delta} - \frac{1}{T}\right) \left(1 - \frac{\delta}{\varepsilon}\right) \rightarrow 2.$$

Thus $\text{LW} / \text{LW}^* \rightarrow 1/2$.

It remains to check regret. Bidder 2 never wins and has realized utility 0. Against bidder 1's bid ε , any winning deviation must bid strictly above ε , while bidder 2's value is at most ε ; hence her best sequence utility is 0. Bidder 1's realized utility is $\sum_t (v_{1t} - \varepsilon)$. The highest competing bid is always $\varepsilon - 2\delta$, so her best sequence can improve her utility by at most $2\delta T = 1/2$, by winning the same items at prices arbitrarily close to $\varepsilon - 2\delta$. Thus her additive regret is at most $1/2$. Since values are pairwise distinct almost surely, Proposition 3 identifies this with regret against the all-Lipschitz benchmark as well. $\square$

Remark 1 (Fixed Lipschitz constants). *The equivalence in Proposition 3 uses the class of all Lipschitz maps with a finite constant allowed to depend on the realized sample path. If instead one fixes a constant L in advance, an arbitrary bid sequence need not be representable by an L -Lipschitz map, especially when two realized values are very close. Therefore the theorem above should not be read as resolving the fixed- L benchmark.*

6 Conclusion

The main theorem gives a complete and sharp solution for the best budget-feasible sequence-of-bids benchmark: the exact asymptotic liquid-welfare loss is $\max\{\gamma, 2\}$. This closes the gap between the known lower bounds for strong benchmarks and the larger upper bound previously proved for the weaker fixed-shading benchmark.

For the broader Fikioris–Tardos open direction, the result is best described as substantial partial progress. It gives the same sharp guarantee for unrestricted Lipschitz value-to-bid maps on distinct sample paths and hence almost surely under independent atomless values. It leaves open, or at least does not address, the fixed- L Lipschitz benchmark and the question of whether stochasticity alone improves guarantees under the original fixed-shading behavioral assumption.

References

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